The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Beyond 2D Video Experiences

Evaluating video formats beyond 2D: stereoscopic, multi-view plus depth, point clouds and dynamic meshes

3GPP

NOVEMBER 2026 – www.5g-mag.com

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About the project

The documents the project rests on

3GPP TR 26.956

Technical Report

Evaluation and Characterization of Beyond 2D Video Formats and Codecs, the output of the study FS_Beyond2D.

3GPP TS 26.501

and TS 26.512

The 5G Media Streaming architecture and protocols the formats are assessed against.

ISO/IEC 23090-5

Standard

Visual Volumetric Video-based Coding (V3C), referenced for evaluation.

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Formats under evaluation

The four format families of the study, with the coding approaches evaluated for each

Stereoscopic video a left and right view: a fixed 3D effect, no free viewpoint

MV-HEVC | Simulcast or frame-compatible HEVC (anchor)

Multi-view plus depth several views with depth: a limited change of viewpoint

MIV profile of V3C | MV-HEVC

Dense dynamic point clouds a volumetric capture of an object or person

V-PCC | G-PCC, noted as a related technology

Dynamic meshes a volumetric representation with time-varying textured 3D meshes

V-DMC

The report also surveys formats under research (neural radiance fields, light fields, 3D Gaussian splatting) without evaluating them.

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Services evaluated

The three services of the study, as listed by the reference repository

Stereoscopic

Live, device to device

UE-to-UE stereoscopic video live streaming. Planned in the repository.

Point cloud

On demand

Streaming of professionally produced volumetric video, a single asset containing people.

Multi-view

Plus depth, on demand

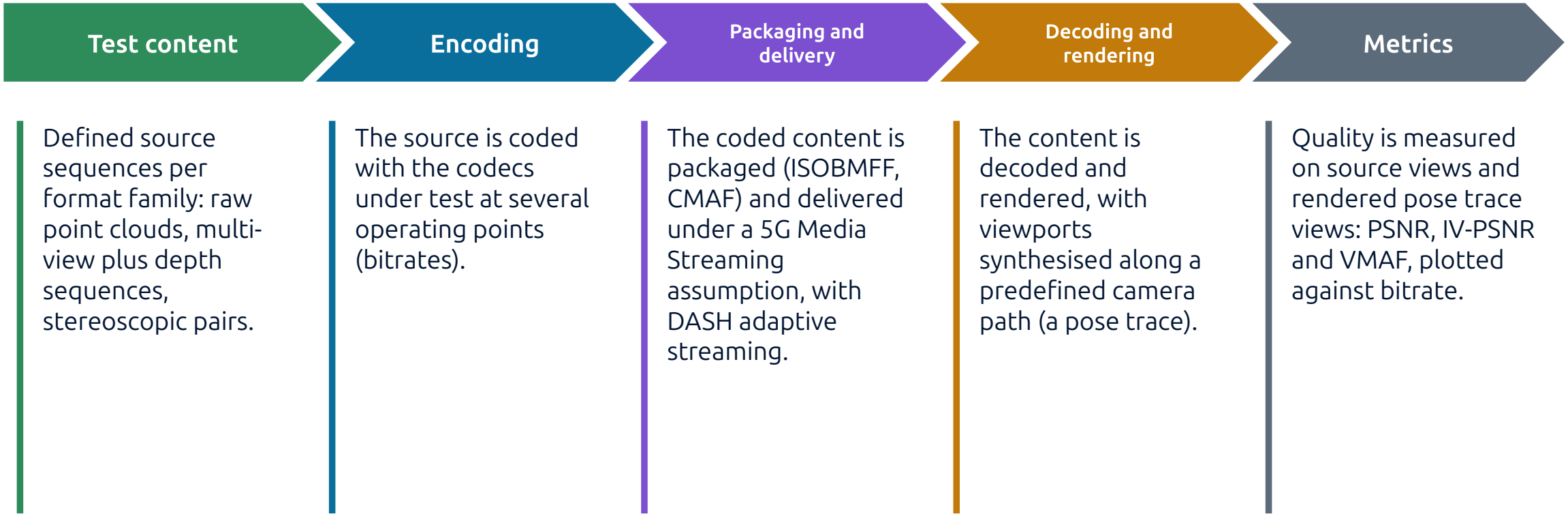
On-demand streaming of multi-view plus depth.

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Evaluation pipeline

A repeatable pipeline, so that results are comparable across formats and contributors

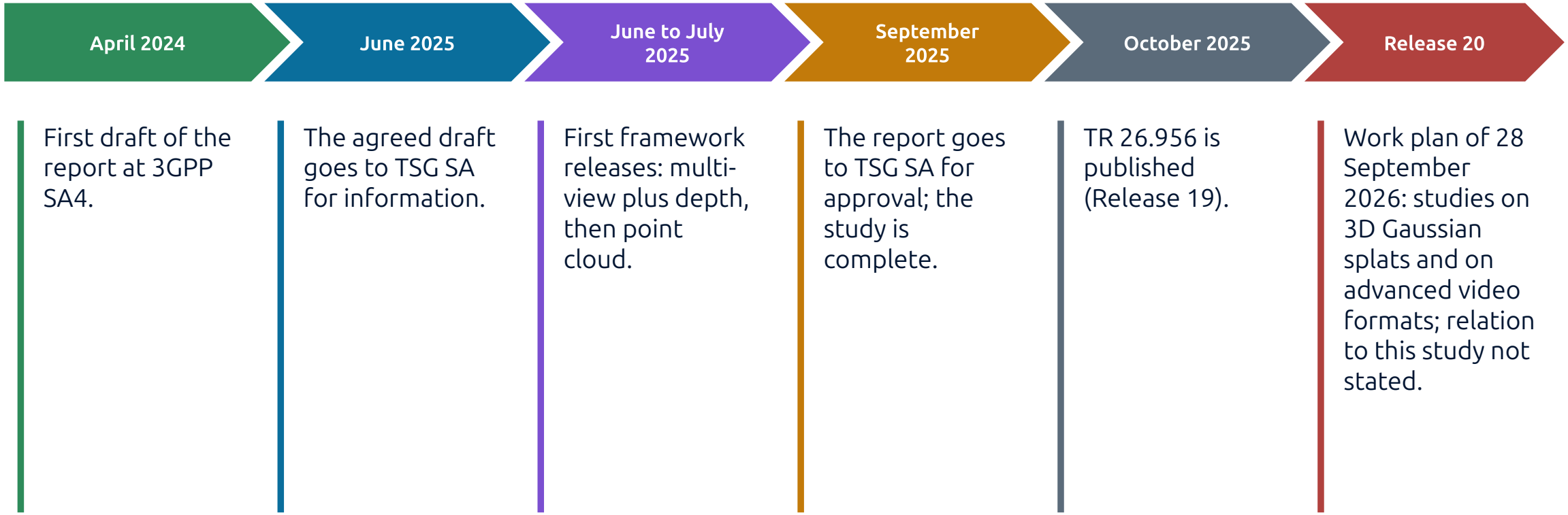


Sources, bitstreams and rendered pose trace videos are large, so the framework references content on external servers.

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How it grew

The 3GPP study and the 5G-MAG evaluation framework, from the first draft to Release 20



Related Release 19 work: VOPS adds TS 26.265, with stereoscopic operation points based on MV-HEVC.

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Repository

The one repository of the evaluation framework, from its README

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-beyond2d-evaluation-framework	scripts and configuration files for the reproducible evaluation of FS_Beyond2D: multi-view plus depth (TMIV) and V-PCC point clouds	Multi-view plus depth, Point cloud	v0.2.1 24 Jul 2025	GitHub

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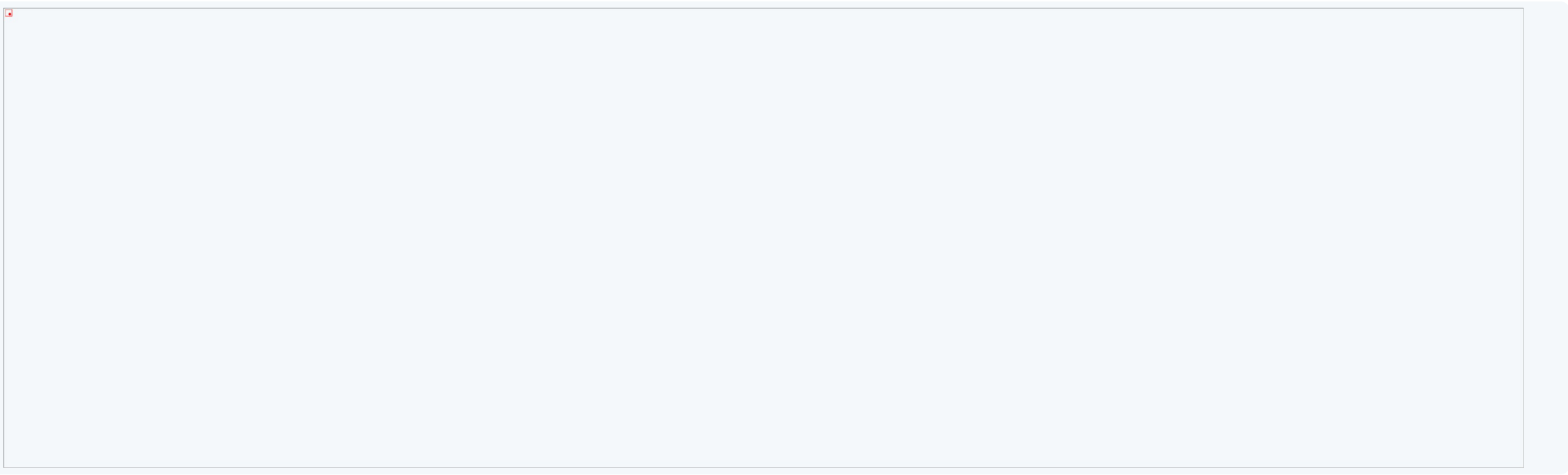
● released ● no release yet

More information at 5g-mag.com/tech/beyond-2d

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **rt-beyond2d-evaluation-framework**

Mapped from the repository's MVD scenario only; boxes without a badge have no repository.

WHERE WE STAND

Beyond 2D Video Experiences

BROUGHT TO STUDY	ON-GOING	OUTCOMES
Scenario 1: UE-to-UE stereoscopic video live streaming	Completing the evaluation setups	Multi-view plus depth evaluation framework
Scenario 2: Streaming of professionally produced volumetric video with a single asset containing people		Point cloud (V-PCC) evaluation scripts for scenario 2
Scenario 3: On-demand streaming of multi-view plus depth		Point cloud setup and video generation scripts

RELATION WITH SDOS AND INDUSTRY FORA 3GPP

3GPP SA4 Feasibility Study on Beyond 2D Video (FS_Beyond2D, SP-240479), which will result in TR 26.956

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